



TANTÁRGYAK

TANANYAGOK

PROJEKTEK

EBR

PRACTICE 01. – WEB PROGRAMMING

1. Create a variable that stores your name. Welcome yourself in the console!
2. In JavaScript you often generate HTML strings. Write to the console!
 - a. Welcome yourself in an `h1` tag!
 - b. The `isLoggedIn` variable shows whether the user is logged in or not. If the user is logged in, generate a welcome message and a logout link. If not, generate a login link. Wrap the generated HTML string with a `div` element.
 - c. An array of names is given. Generate an unordered list from it!
 - d. An array of student data is given. Student data stores the Neptun id, the name and the date of birth of the student. Generate an HTML table from this array!
3. Write a function that calculates the factorial of an integer!
4. Find a negative number in a series of numbers!
5. Count the even numbers in a series of numbers!
6. An array of numbers is given. Put the prime numbers in a new array!
7. A movie catalogue contains the following informations on every movie:
 - title
 - length
 - year
 - directors
 - actorsCreate a sample catalogue with some dummy data! Solve the following tasks!
 - a. List the movies to the console!
 - b. Return those movies which have several directors (not one)!
 - c. Which is the longest movie?
8. Two numbers are given. Write a function that returns their greatest common divisor! Pseudo-code:

```

Function greatestCommonDivisor(a, b: Integer): Integer
  If a < b then swap(a, b)
  residual = a mod b
  While residual > 0
    a := b
    b := residual
    residual := a mod b
  End of while
  greatestCommonDivisor := b
End of function

```

9. Two numbers are given. Write a function that returns the least common multiple of them! Pseudo-code:

```

Function leastCommonMultiple(a, b: Integer): Integer
  x, y := a, b
  While x ≠ y
    If x < y then x := x + a
    Else If x > y then y := y + b
  End of while
  leastCommonMultiple := x
End of function

```

10. Select the numbers that differ from both neighbors within a predetermined value!
11. Decide whether all of the elements of a matrix are even numbers!
12. During a given period, each day we measured the temperature of the lake Balaton and stored in an array. Solve the following tasks!
- How much was the average temperature of the water during this period?
 - How many times did we measure more than 26 degrees Celsius?
 - Which was the highest measured temperature value?
 - In which day was the coldest water?
 - What day was the water exactly 23 degrees?
 - Occurred that two consecutive days of water temperature changed 3 degrees?
 - How many days did the longest period consist of when the water was over 25 degrees?
13. For a couple of days, we measured the air temperature every noon. Solve the following tasks!
- Select the values when it was freezing!

- At the end of each temperature, put the `C` text!
- What is the highest temperature value!
- How many times the temperature went below 20 degrees!
- Decide whether there is a value over 40 degrees!
- Decide whether each temperature value is positive!
- Enter the first value when the temperature went up to 10 degrees!